

18. Candle-Based Entry Engine (Locked Strike Architecture)

Objective

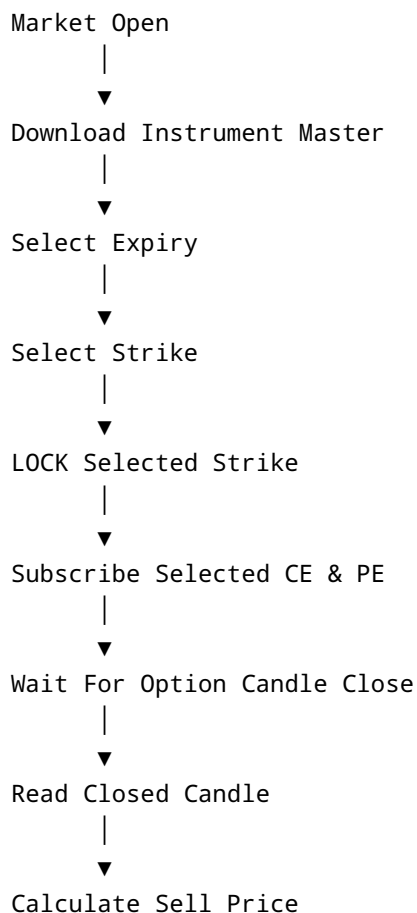
The strategy **must NOT place SELL orders using Live Tick, Bid, Ask, or LTP.**

After the expiry and strike are selected, the strike is **locked** for the entire trade lifecycle.

The engine waits for the **selected option contract's candle** to close and calculates the SELL price from the candle's **Close** price.

This creates deterministic entries and avoids continuous price chasing.

Architecture Flow





Strike Selection

Strike is selected only once.

Example

NIFTY

25135

↓

ATM

25150

↓

ATM Offset = 2

↓

Selected CE

25250

↓

Selected PE

25050

After selection

LOCK STRIKE = TRUE

Until

Position Closed

The strategy **must never change the strike while a position or pending order exists.**

Candle Source

The candle is created using the **selected option contract**, not the NIFTY index.

Example

Instrument

NIFTY

31 JUL

25250 CE

Candle

Time

10:15

Open

12.10

High

12.60

Low

11.90

Close

12.40

The Close price becomes the reference price.

Sell Price Calculation

Formula

SELL PRICE

=

Closed Candle Close

+

Configured Offset

Example

Close

12.40
Offset
1.00
↓
SELL
13.40

PE Example

Closed Candle

Open
11.50
High
11.90
Low
11.20
Close
11.80

Configured Offset

1.00

Calculated

SELL

12.80

Pending Order Logic

If SELL order is not executed

Pending > 10 Seconds

↓

Cancel Pending Order

↓

Do NOT modify order every tick

↓

Wait For Next Candle Close

↓

Read New Close

↓

Calculate New Sell Price

↓

Place New SELL Order

The engine must never continuously chase market prices.

Filled Order Flow

SELL Filled

↓

Immediately Create BUY Order

Formula

BUY TARGET

=

SELL PRICE

-

TARGET

Example

SELL

13.40

Target

1

↓

BUY

12.40

Stop Loss

If premium rises

SELL

13.40

↓

Premium

15.40

↓

SL Hit

↓

Market BUY

↓

Trade Closed

Position Timeout

If neither Target nor Stop Loss is reached

Trade Open

↓

Position Timeout

↓

Exit Position

↓

Reason

TIMEOUT

Candle Timeframes

Supported

15 Seconds

30 Seconds

1 Minute

3 Minutes

5 Minutes

Recommended

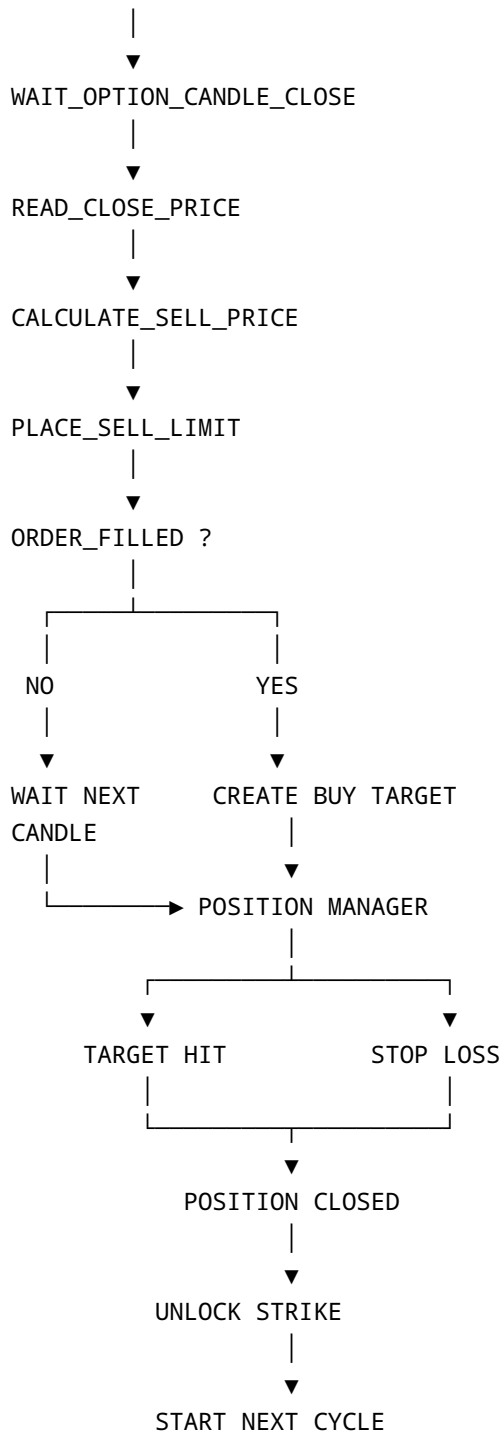
1 Minute

Configuration

```
{
  "entryMode": "OPTION_CANDLE_CLOSE",
  "priceSource": "CLOSE",
  "lockStrike": true,
  "reQuoteOnNextCandle": true,
  "sellOffset": 1.0,
  "target": 1.0,
  "stopLoss": 2.0,
  "pendingTimeout": 10,
  "positionTimeout": 60,
  "candleTimeframe": "1m"
}
```

State Machine

```
SELECT_EXPIRY
  |
  ▼
SELECT_STRIKE
  |
  ▼
LOCK_STRIKE
```



Engineering Rules

1. Strike is selected only once per trade cycle.
2. Strike remains locked until the position is completely closed.

3. SELL price is always calculated using the **closed option candle**.
4. Never use Live Bid, Live Ask, or LTP for the initial SELL calculation.
5. Pending SELL orders are cancelled only after timeout and recalculated using the **next completed candle**.
6. No tick-by-tick order modification.
7. No strike switching while a position or pending order exists.
8. Every state transition must be logged with timestamps and exit reasons.